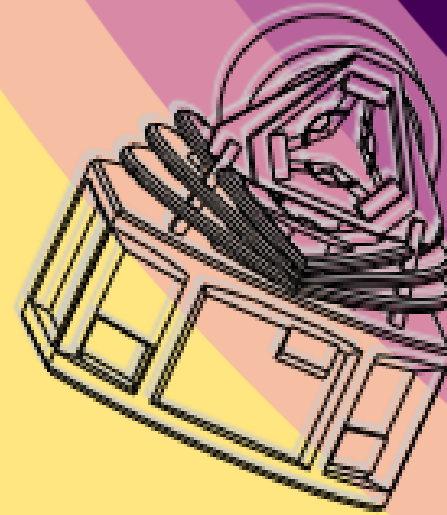


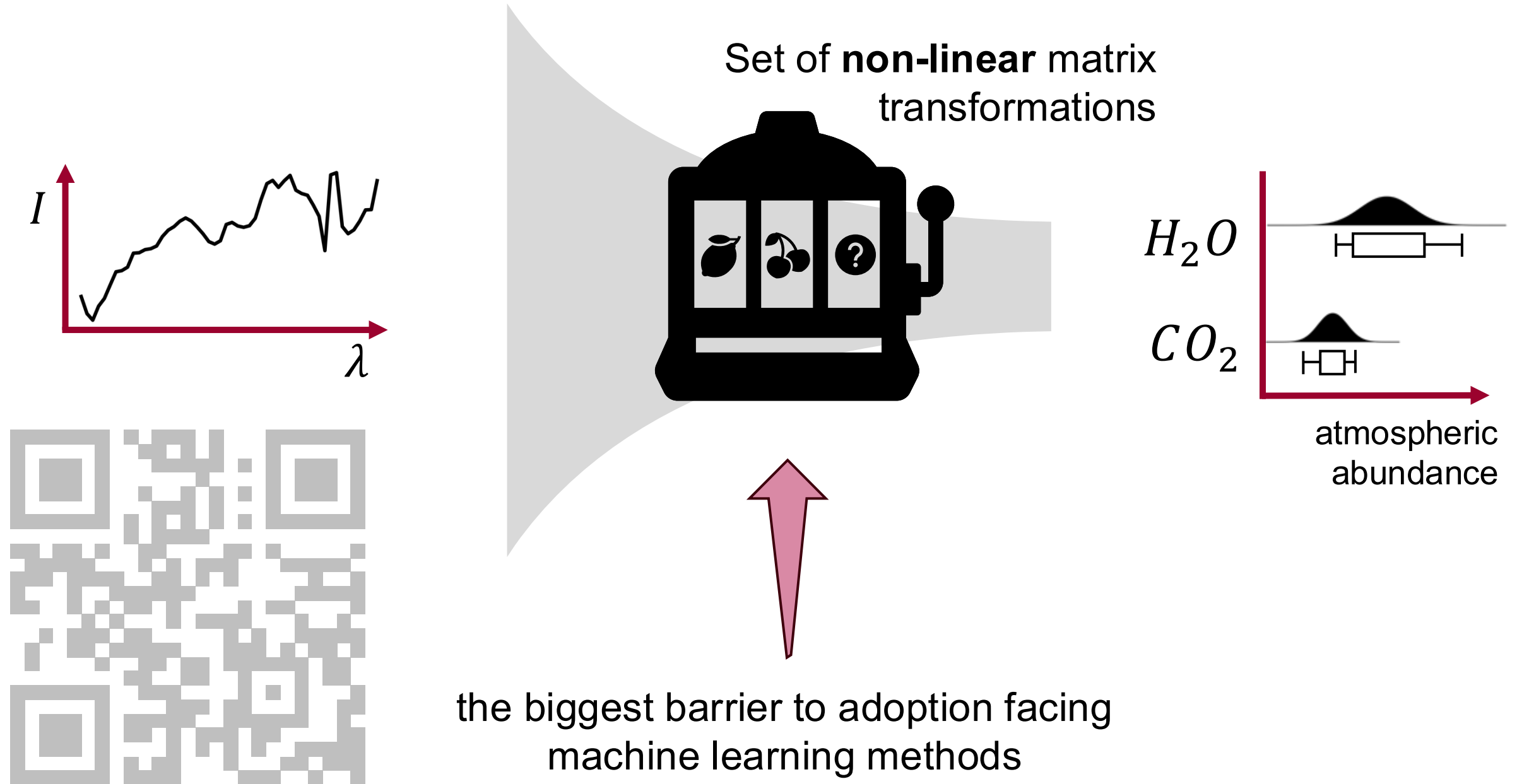
Interpreting and Validating Interpretability Metrics

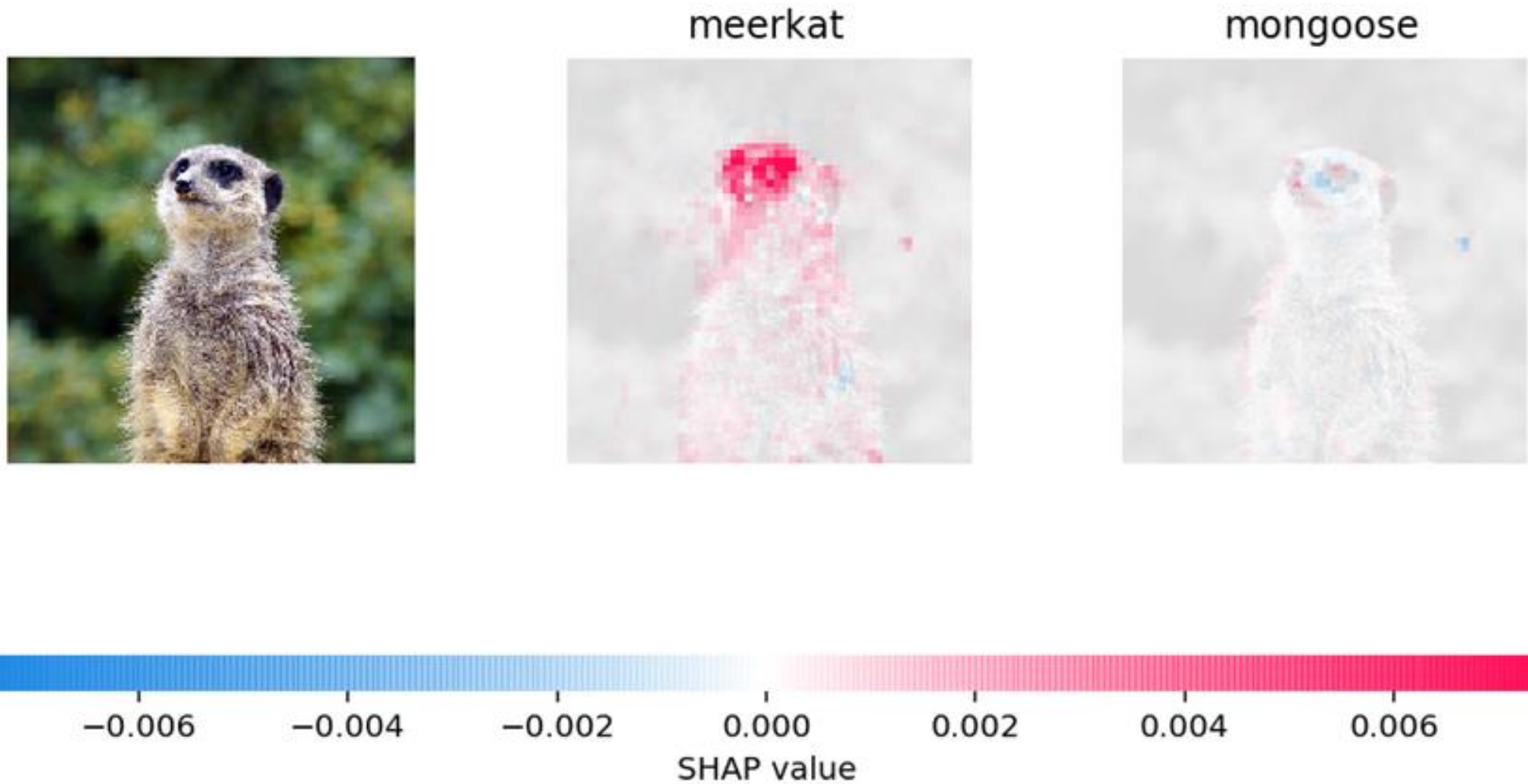
Jools Clarke

Contact me: joolsclarke.co.uk / j.d.clarke@ucl.ac.uk

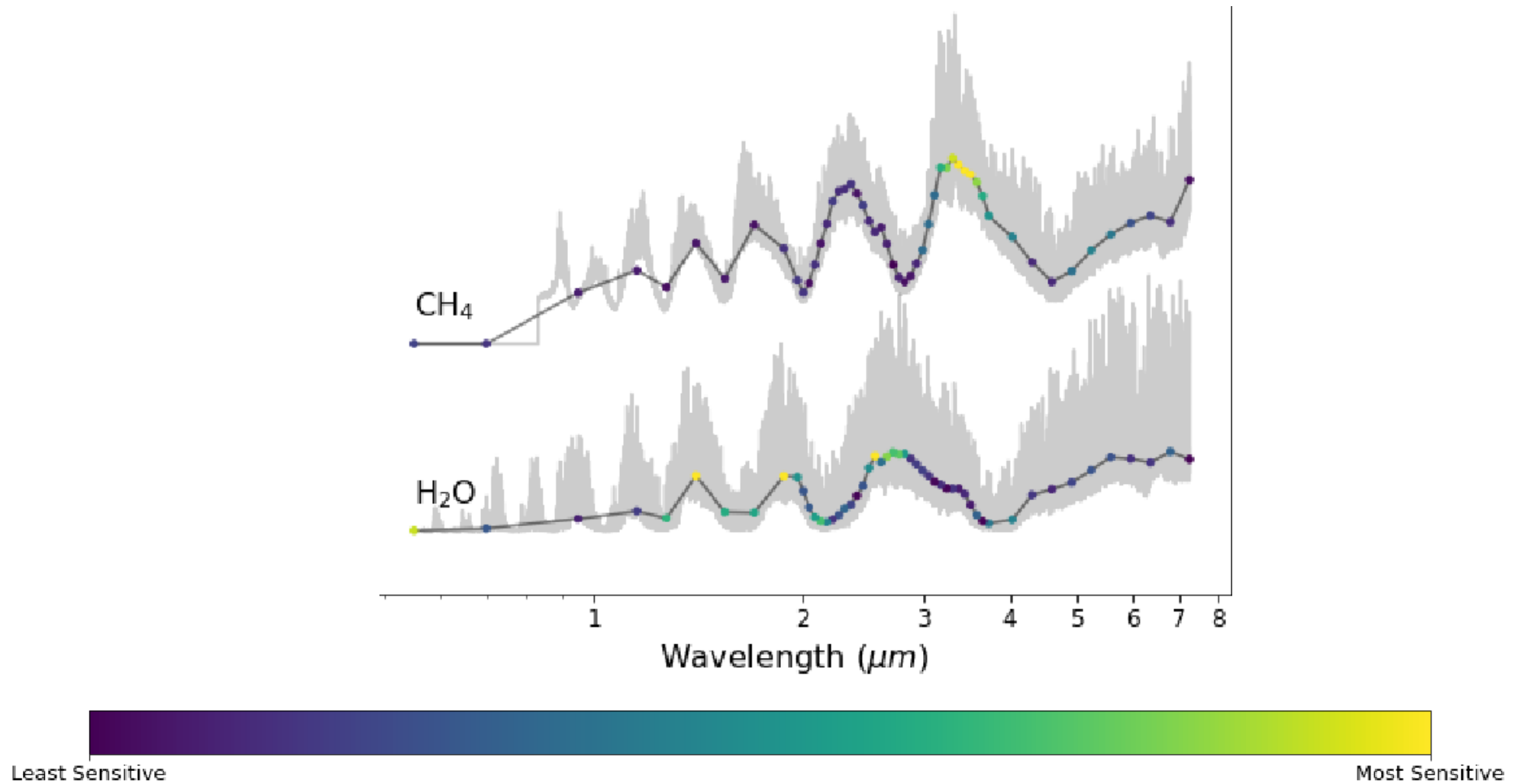
Download these
slides



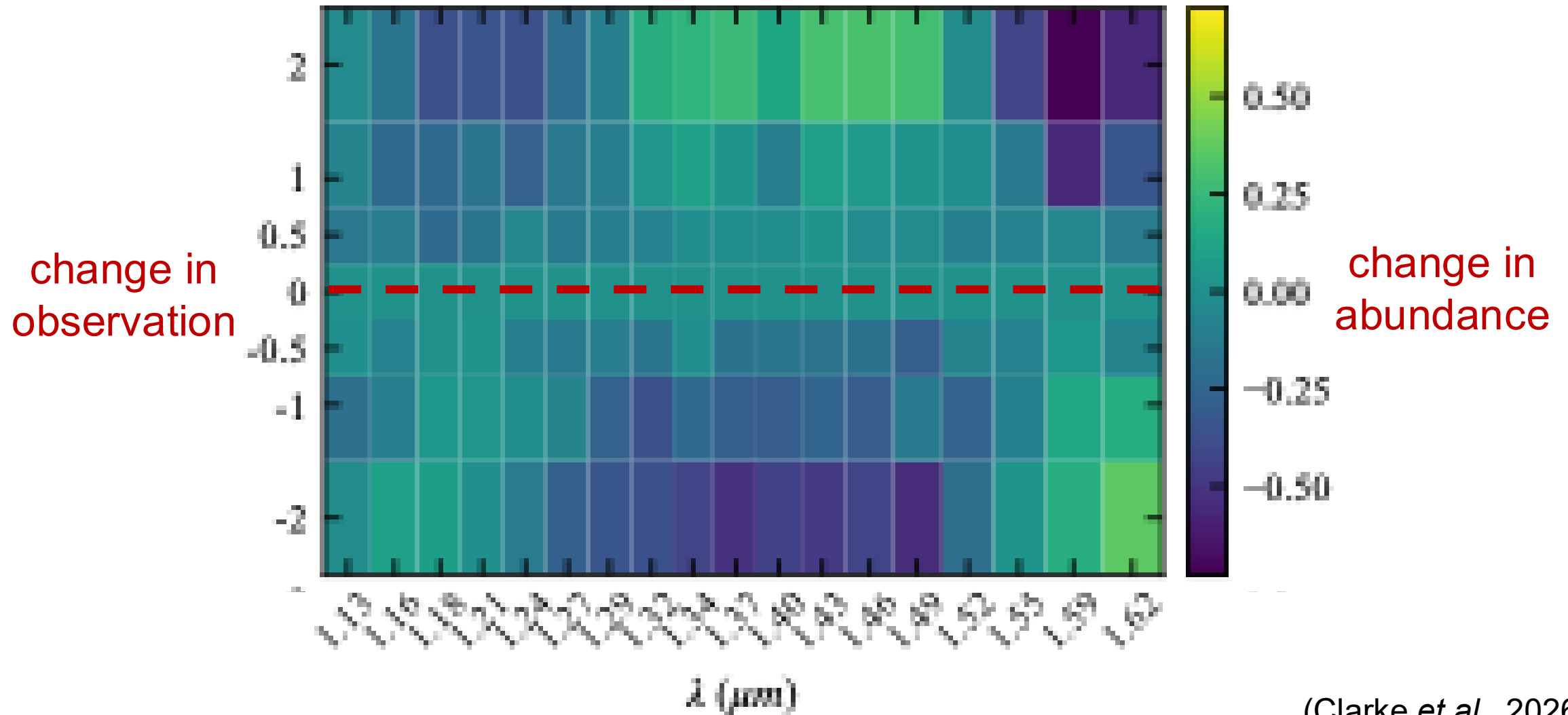


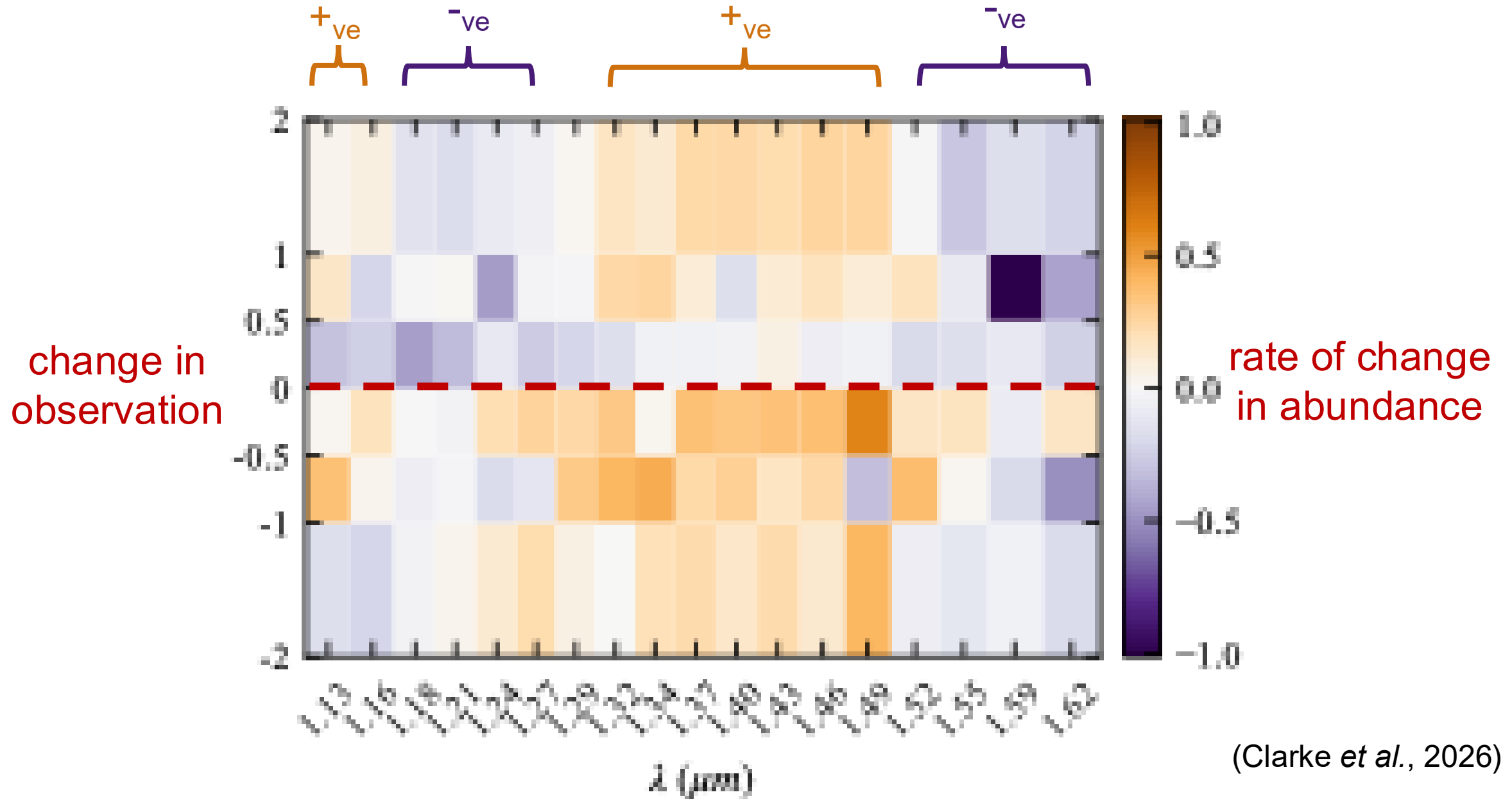


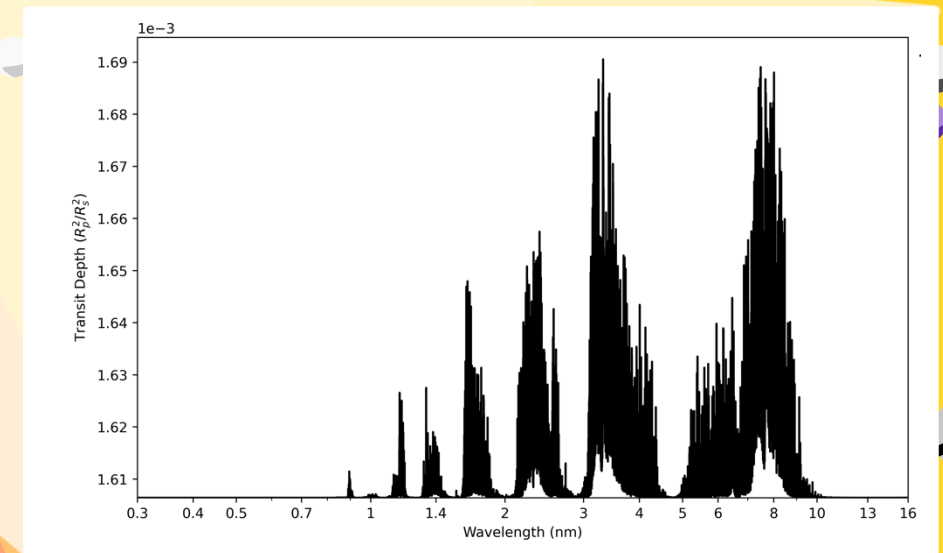
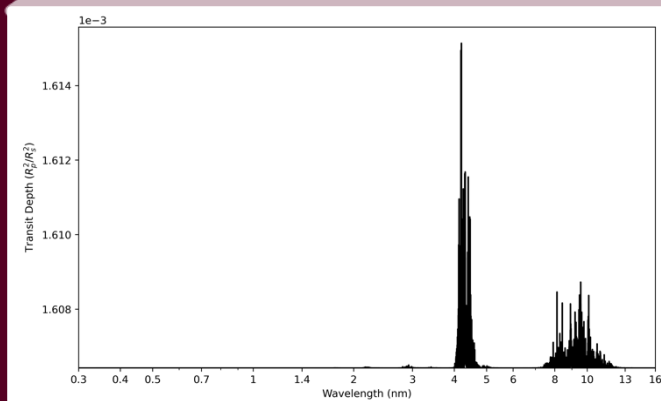
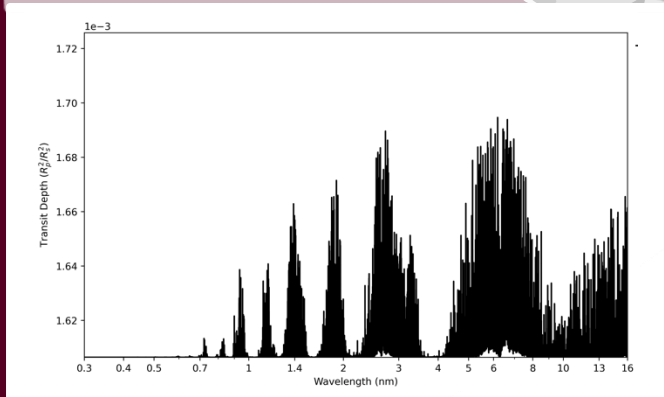
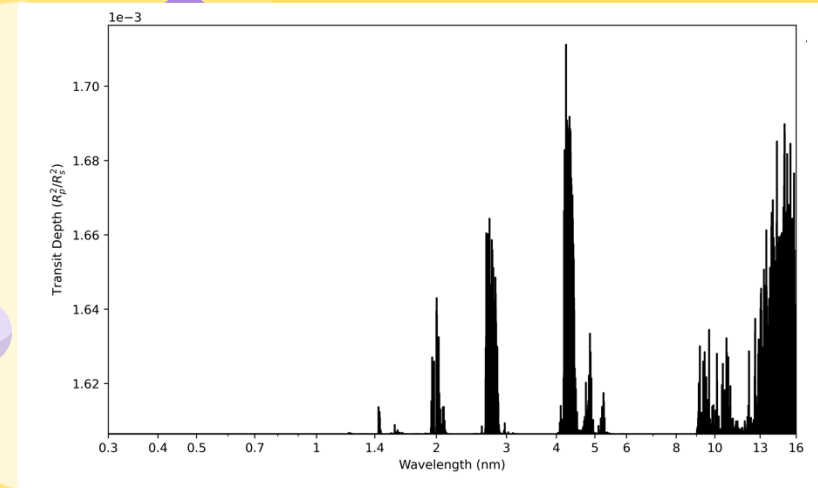
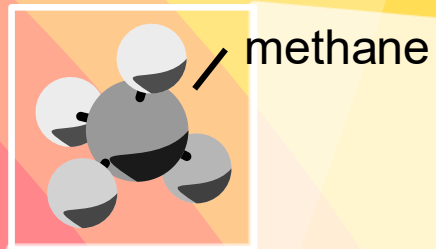
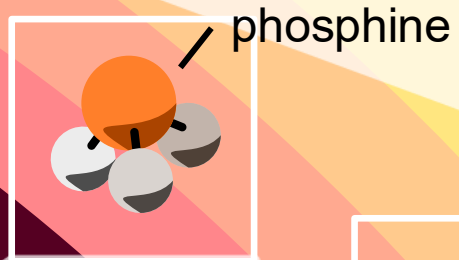
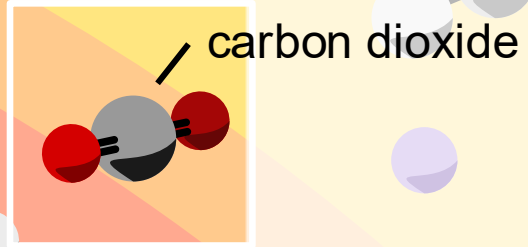
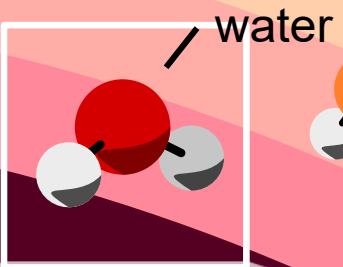
SHAP paper: (Lundberg and Lee, 2017), image: (Shojaeinasab *et al.*, 2024)

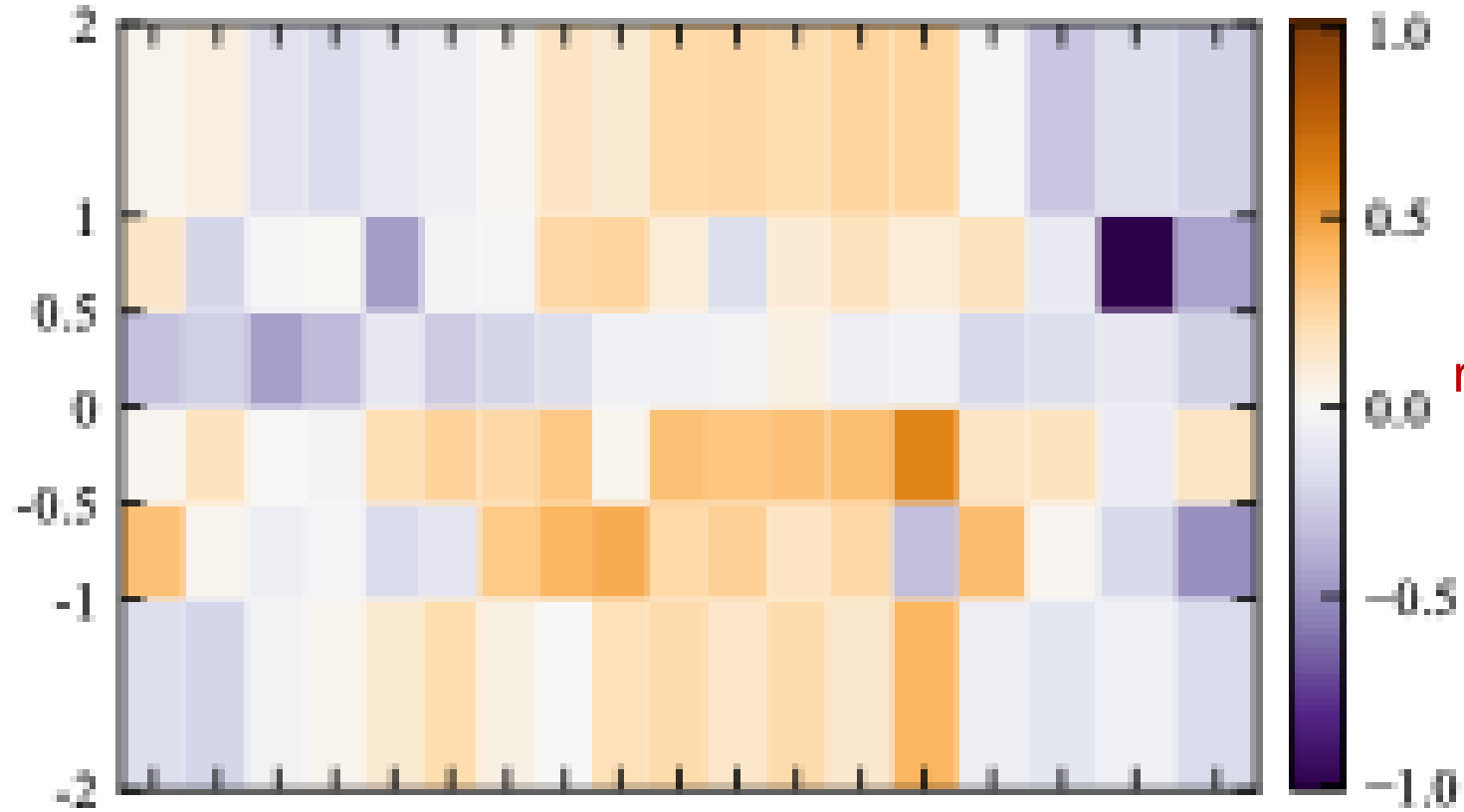
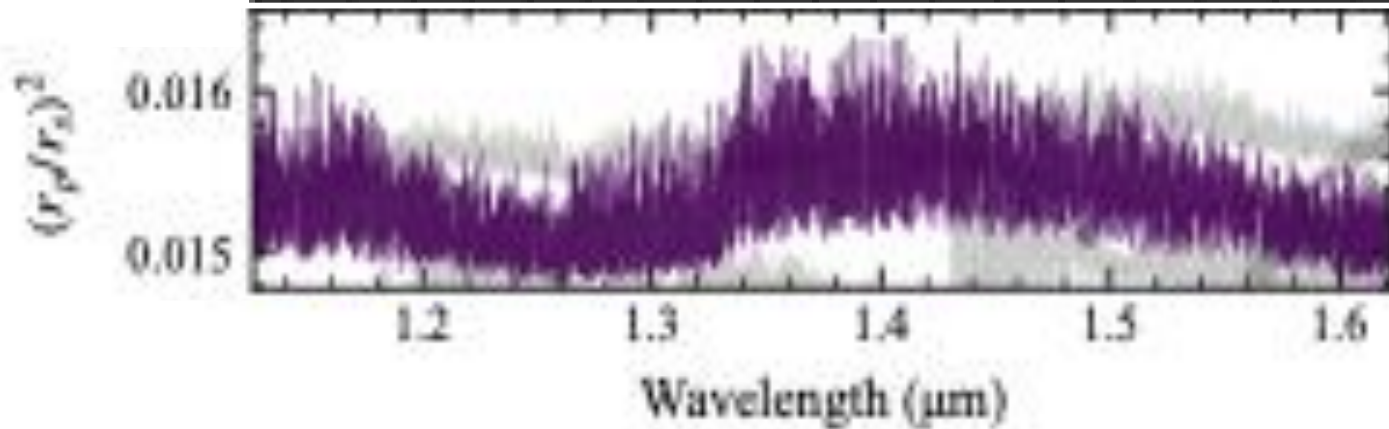


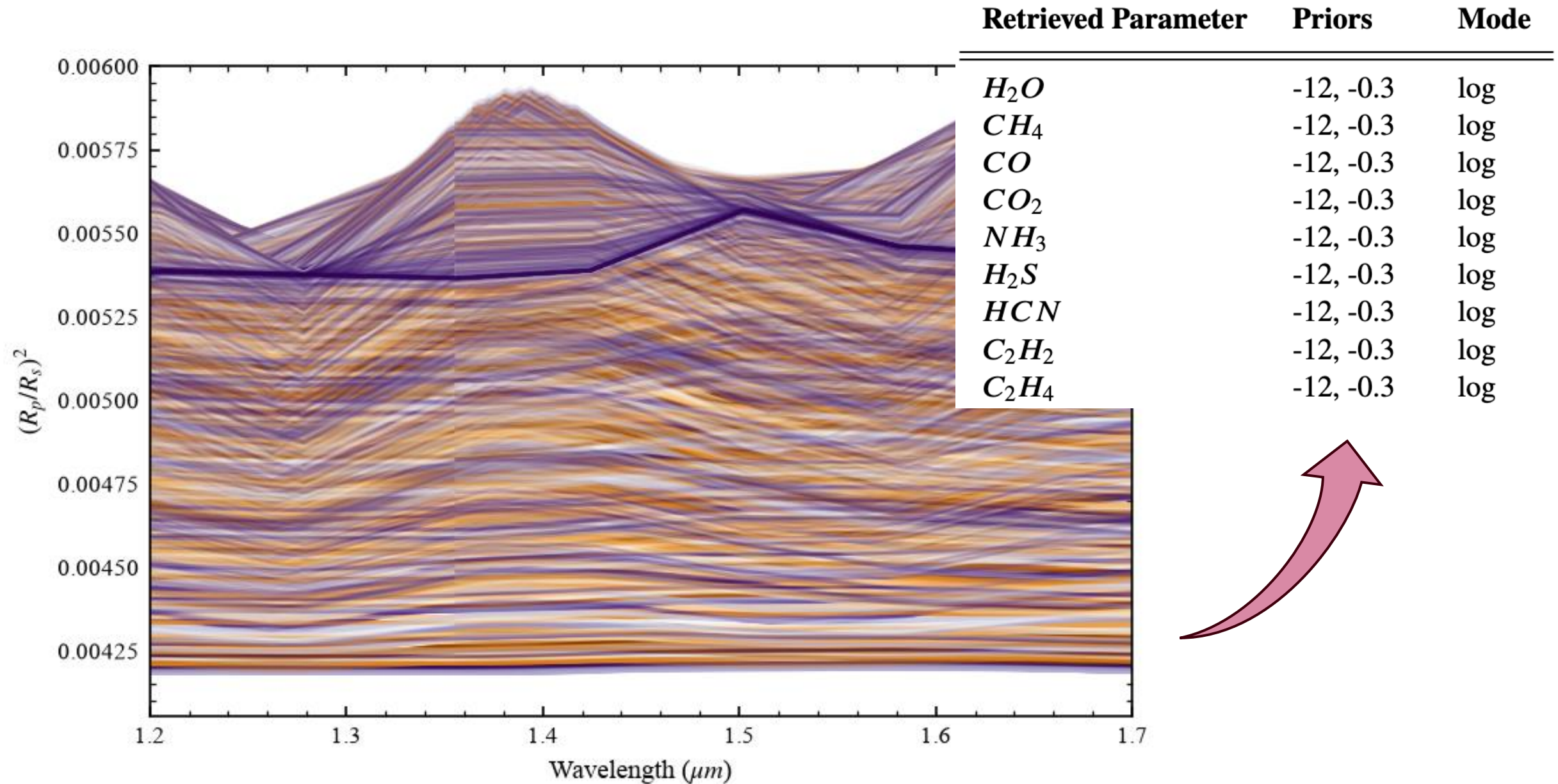
(Yip *et al.*, 2021)



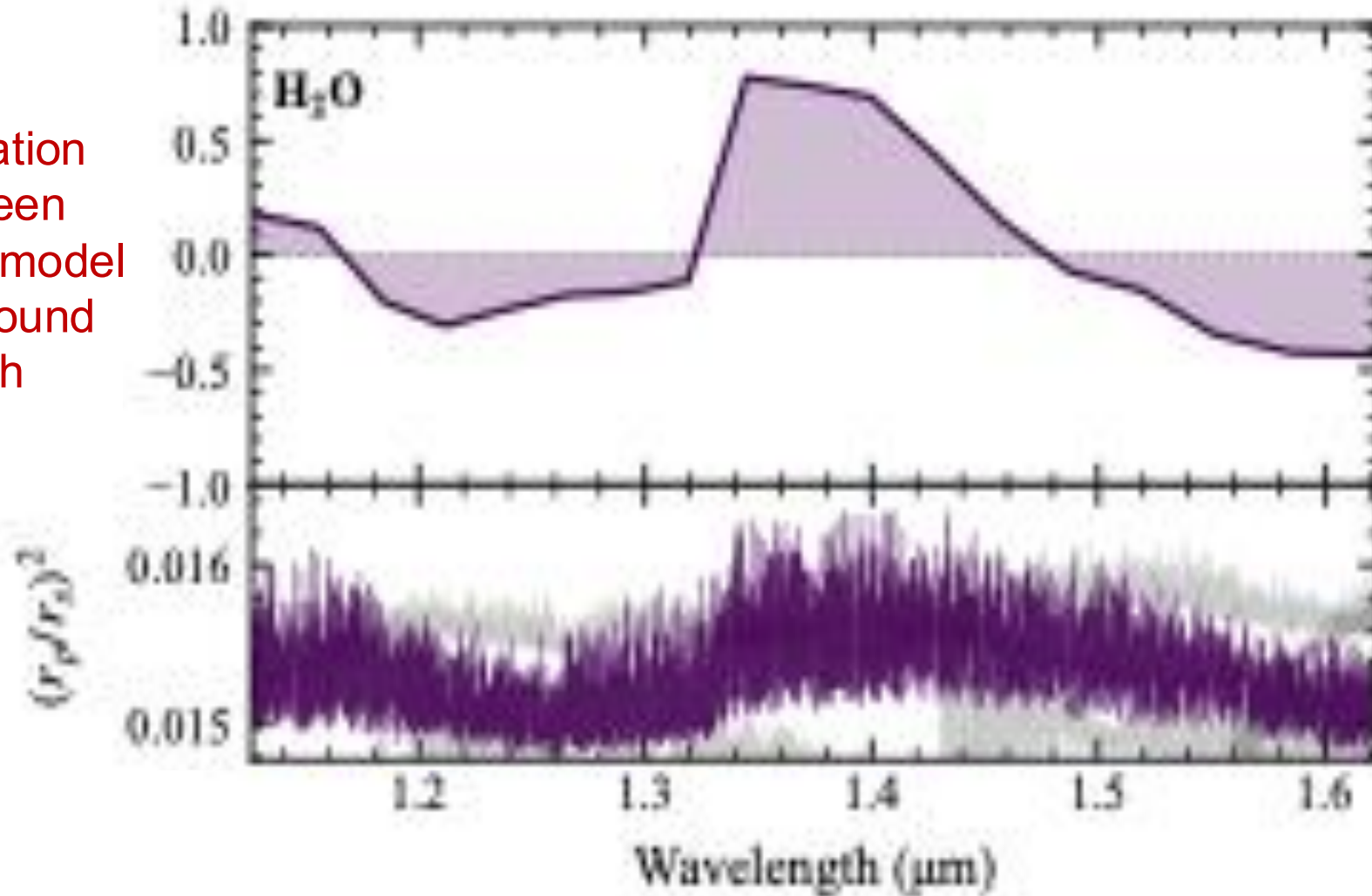


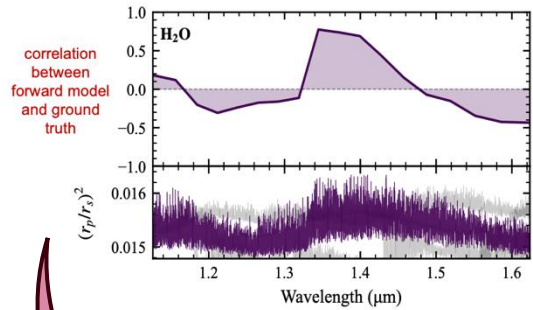


change in
observationrate of change
in abundancewater
contribution

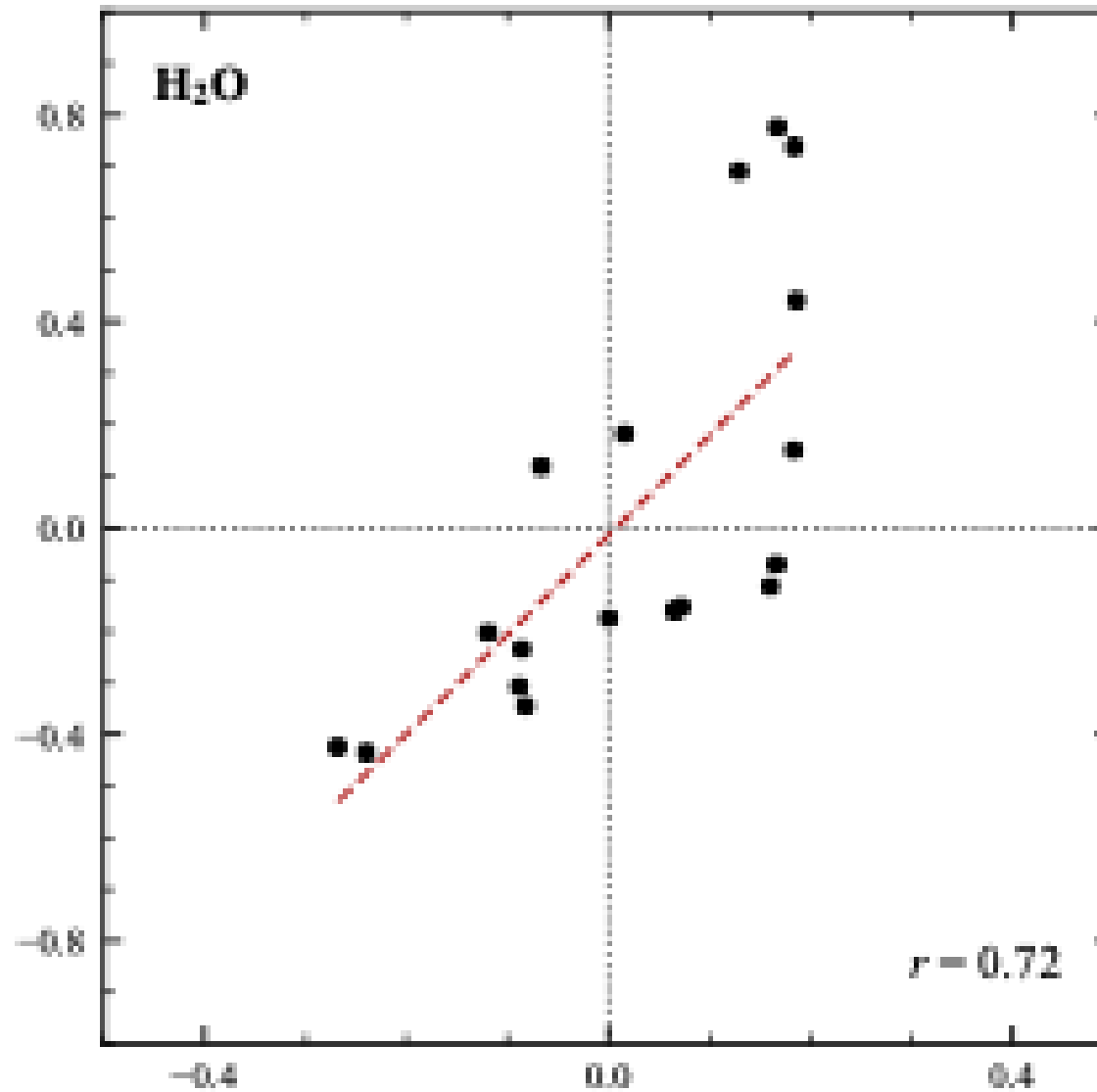


correlation
between
forward model
and ground
truth



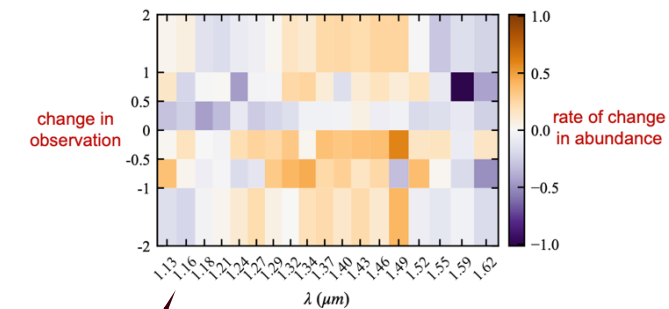


correlation between forward model and ground truth

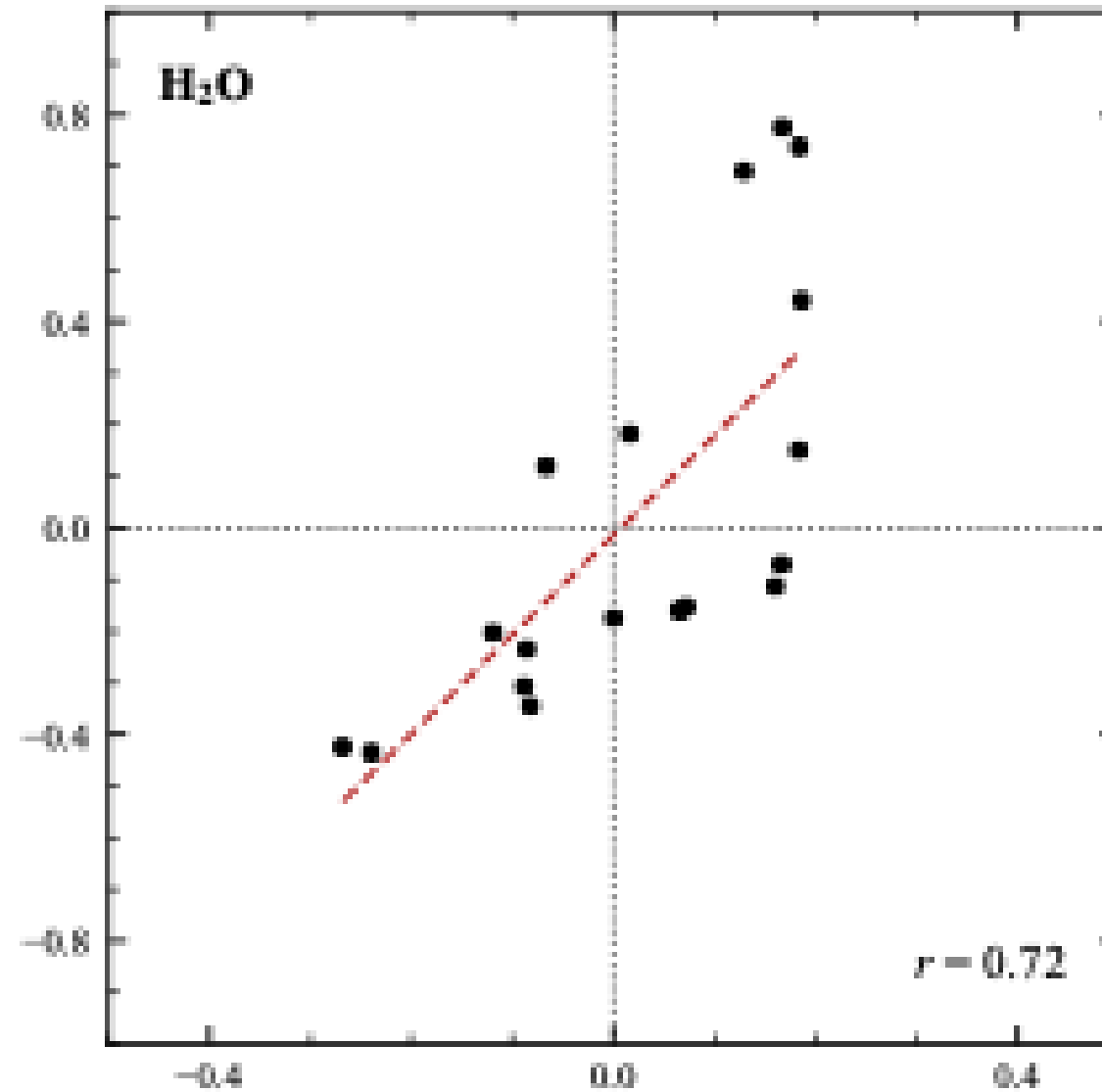


average rate of change in abundance

1 point
=
1 wavelength



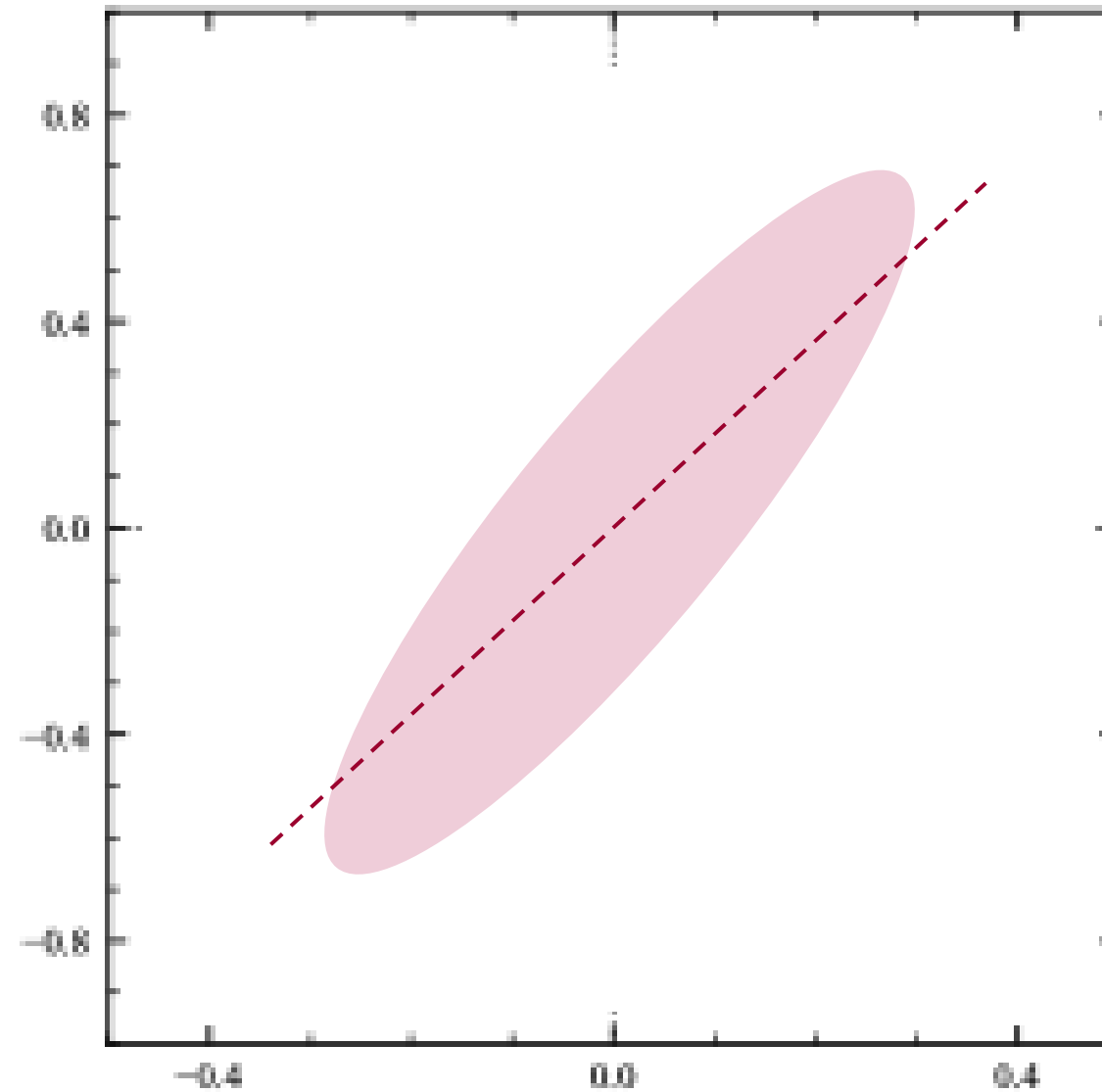
correlation between
forward model
produced by **TauREx-III**
and ground truth



**What is a good
metric?**

average rate of change in mean posterior
abundance **retrieved by TauREx-III**

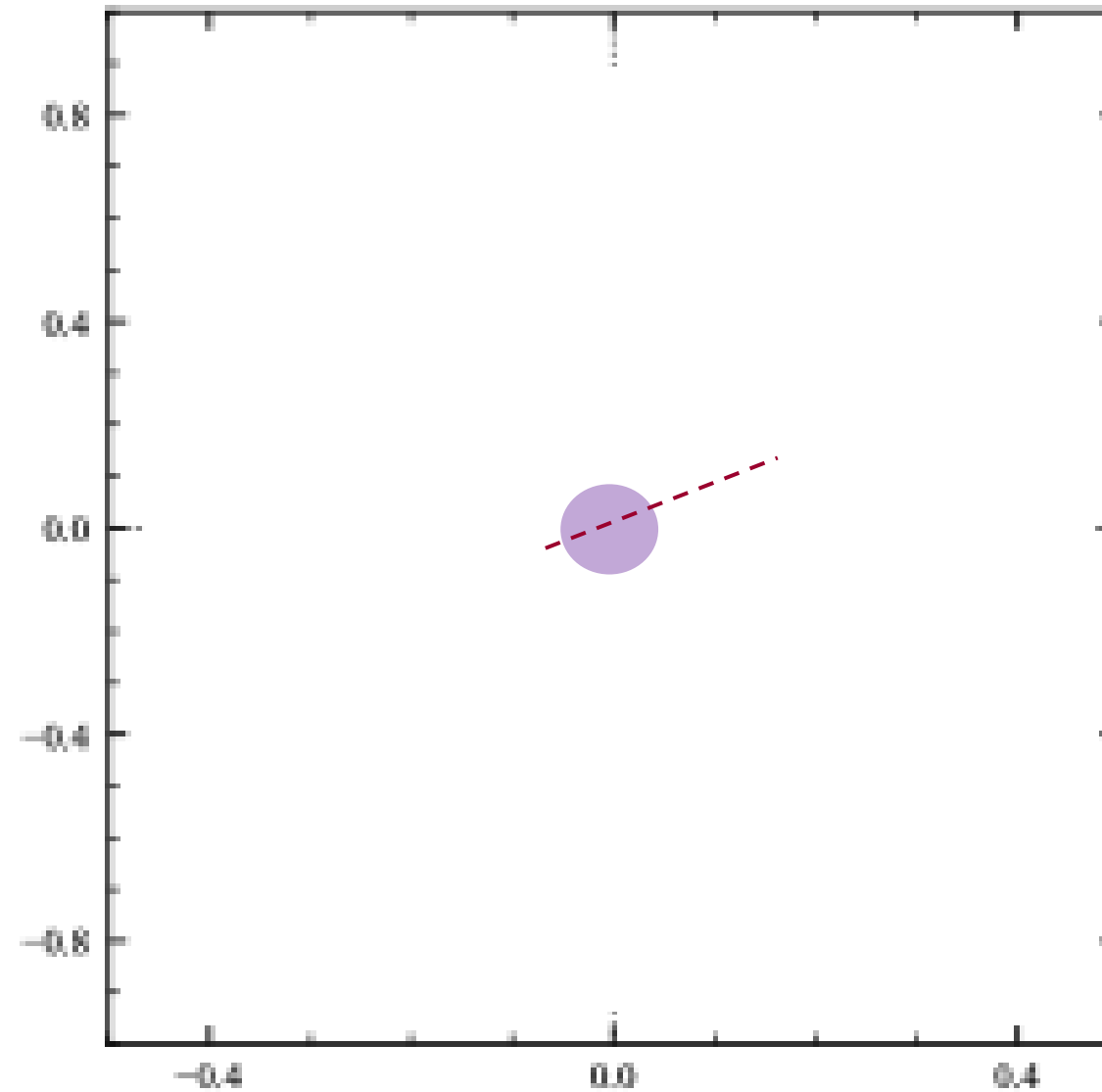
average rate of change in
mean posterior abundance
retrieved by ML surrogate



average rate of change in mean posterior
abundance retrieved by TauREx-III

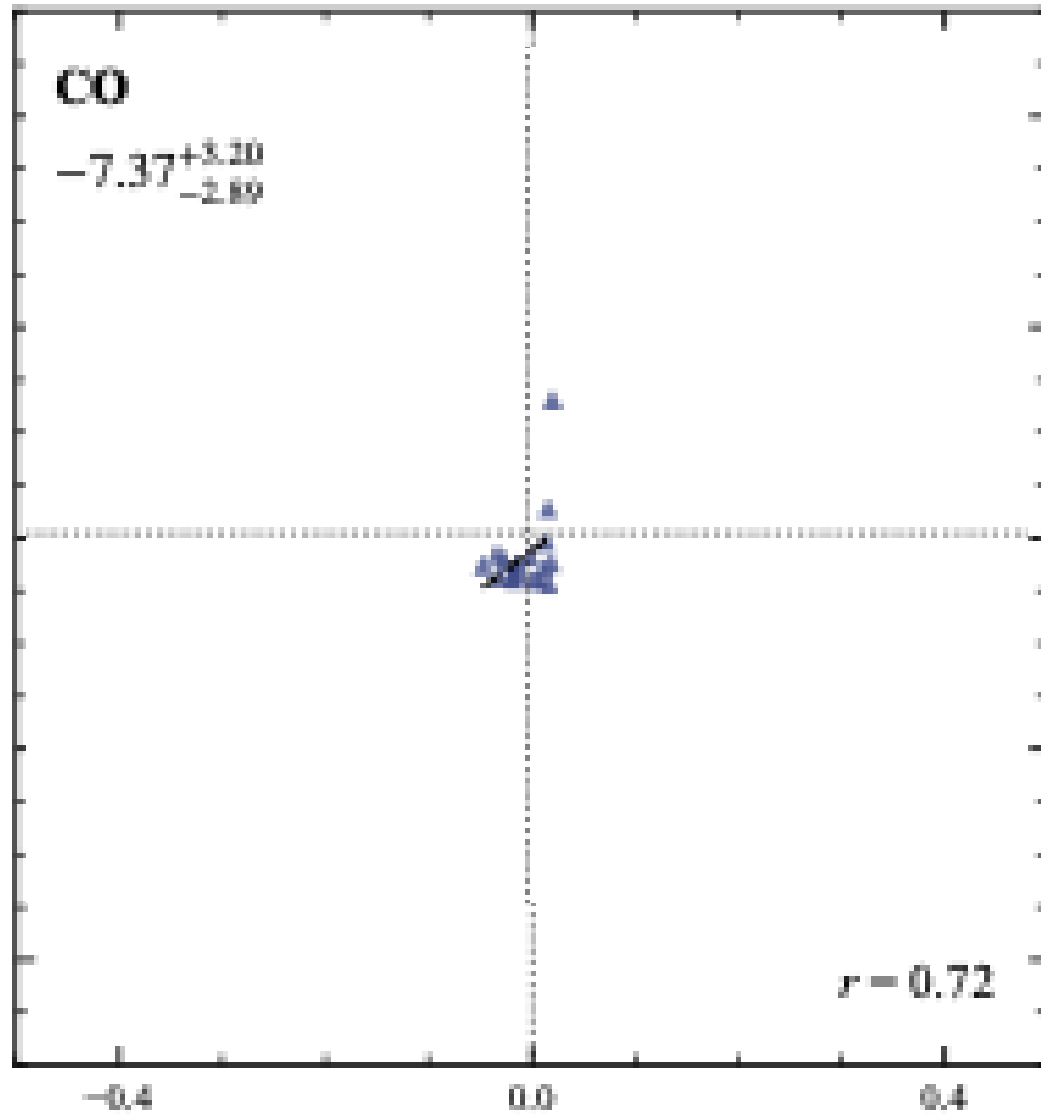
The ML surrogate
interprets the data
in *the same way* as
TauREx.

average rate of change in
mean posterior abundance
retrieved by ML surrogate



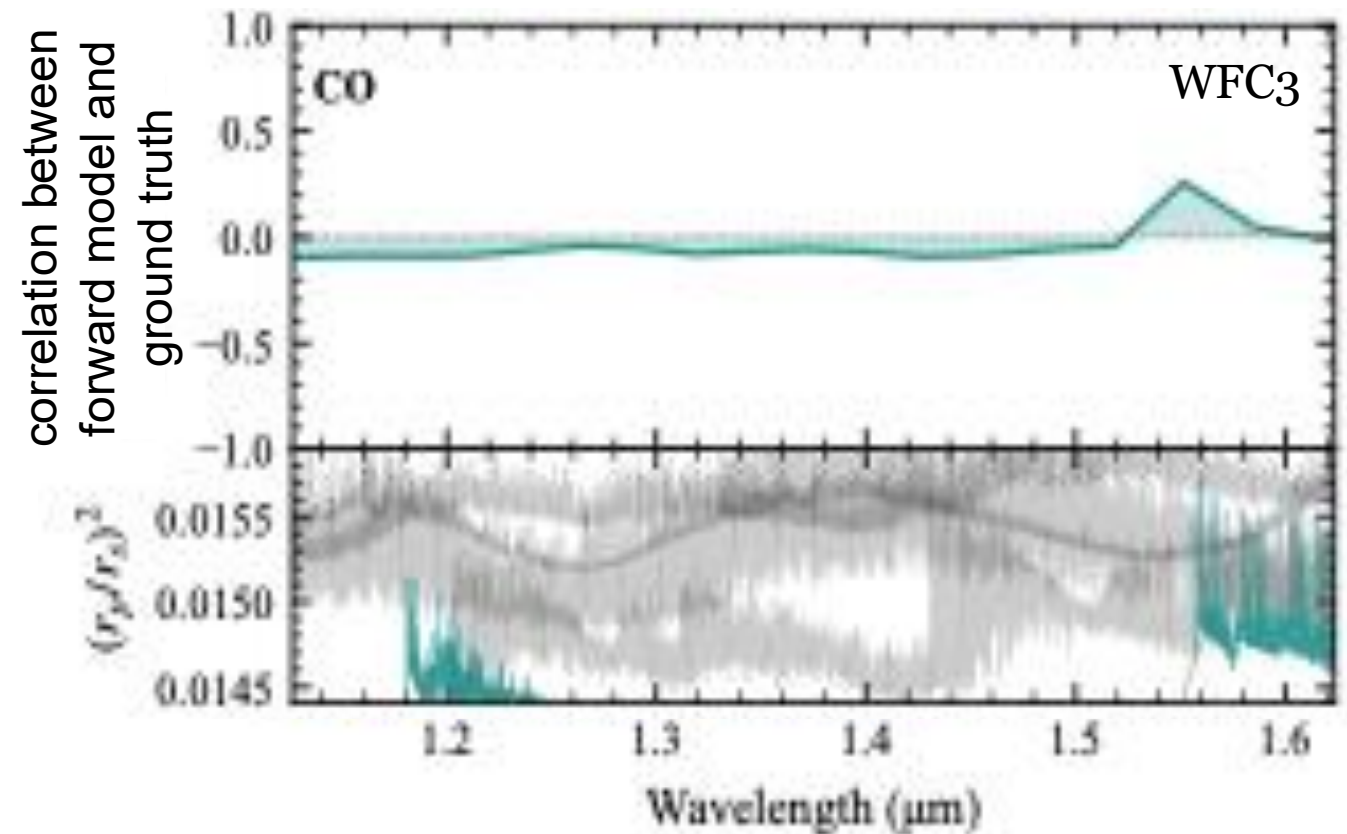
**The data does not
constrain the
solution well.**

average rate of change in mean posterior
abundance retrieved by TauREx-III

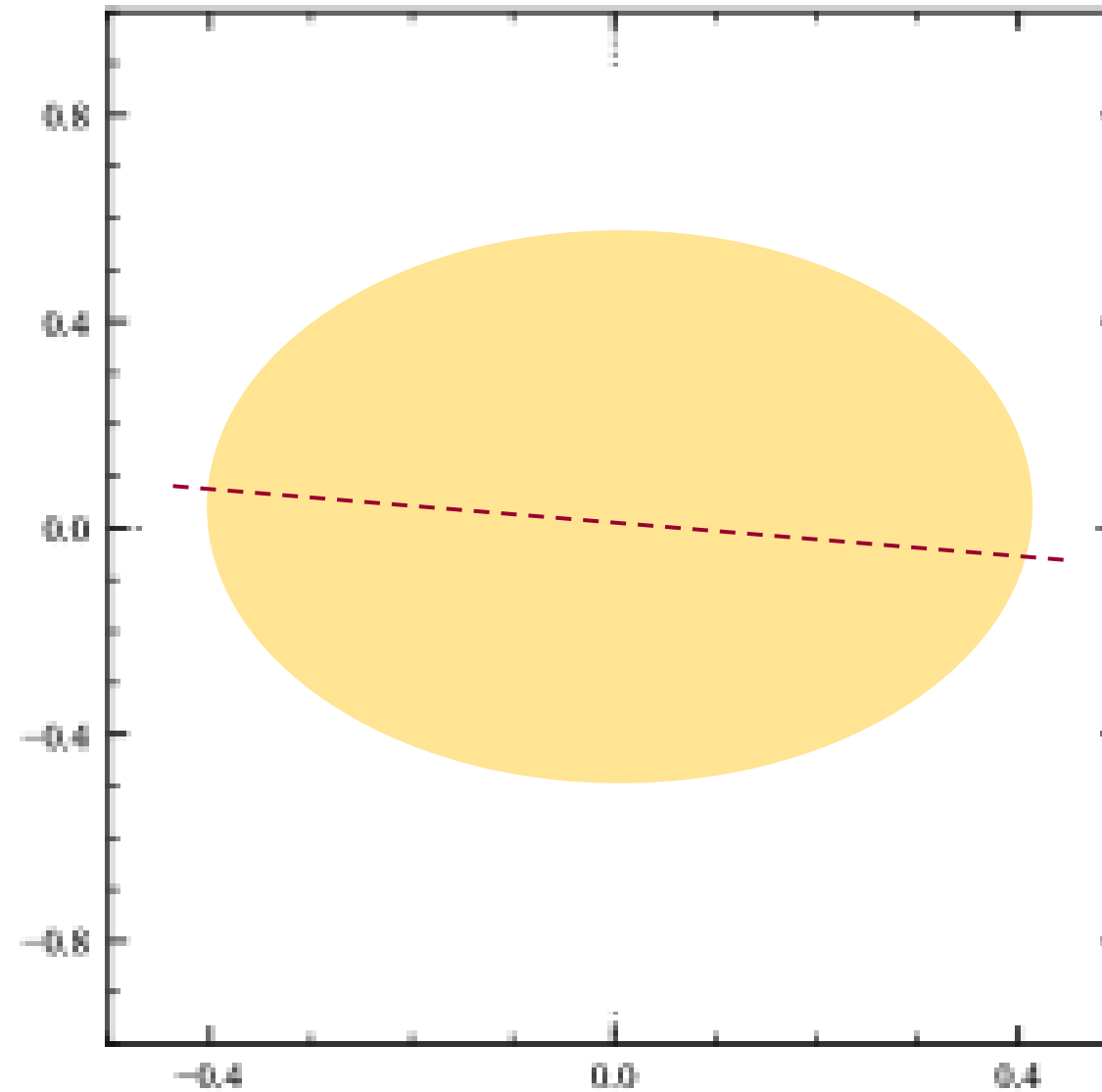


average rate of change in mean posterior
abundance **retrieved by TauREx-III**

The data does not
constrain the
abundance well.



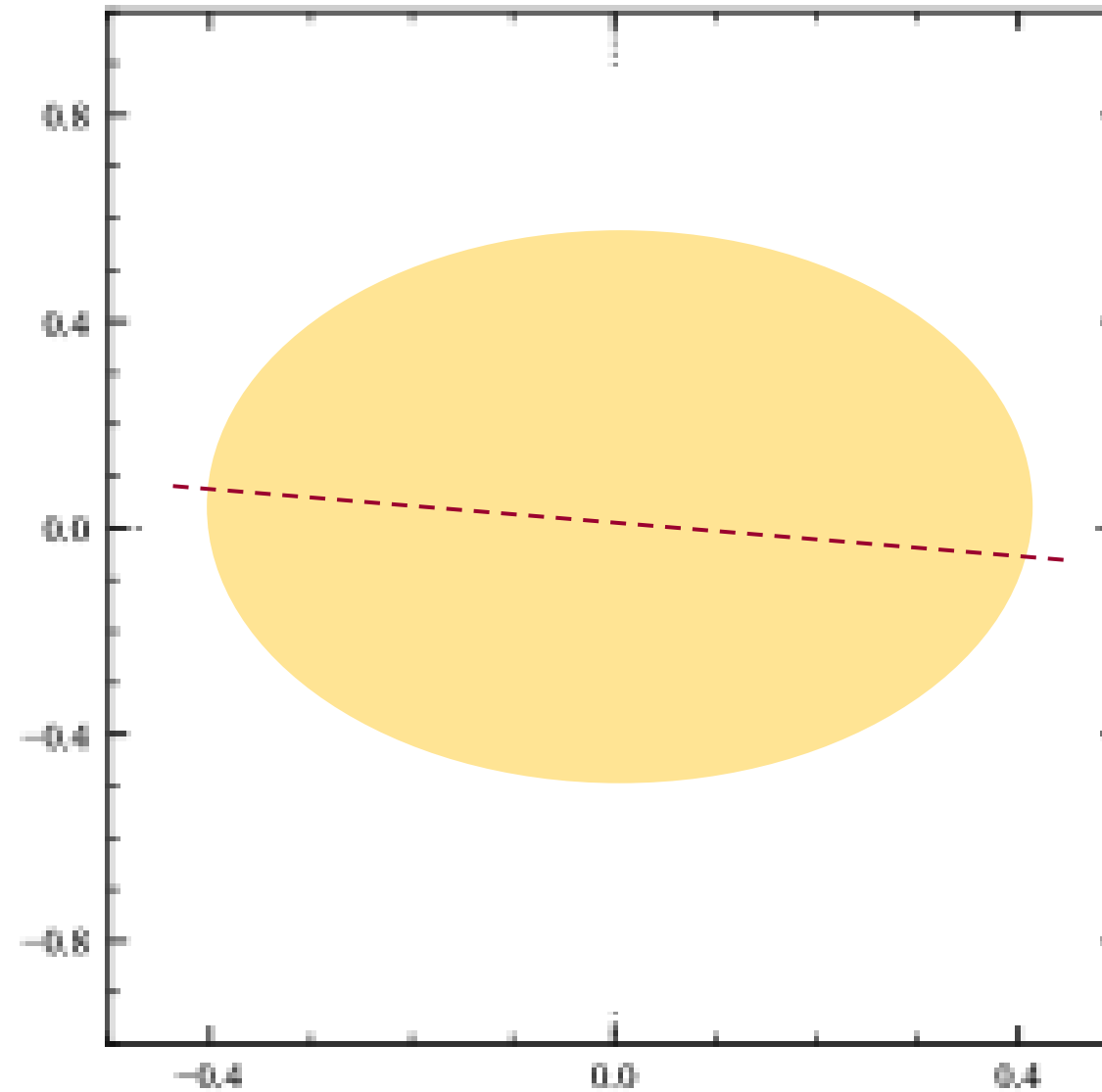
average rate of change in
mean posterior abundance
retrieved by ML surrogate



The ML surrogate
interprets the data
differently to
TauREx.

This is *not* the same
as low model
accuracy.

average rate of change in
mean posterior abundance
retrieved by

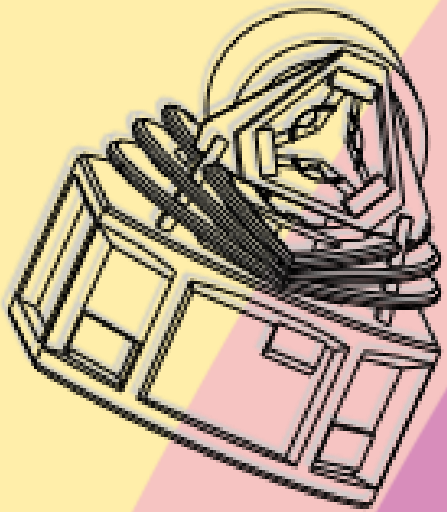


**Poor model
optimisation?**

**New ways to break
degeneracy?**

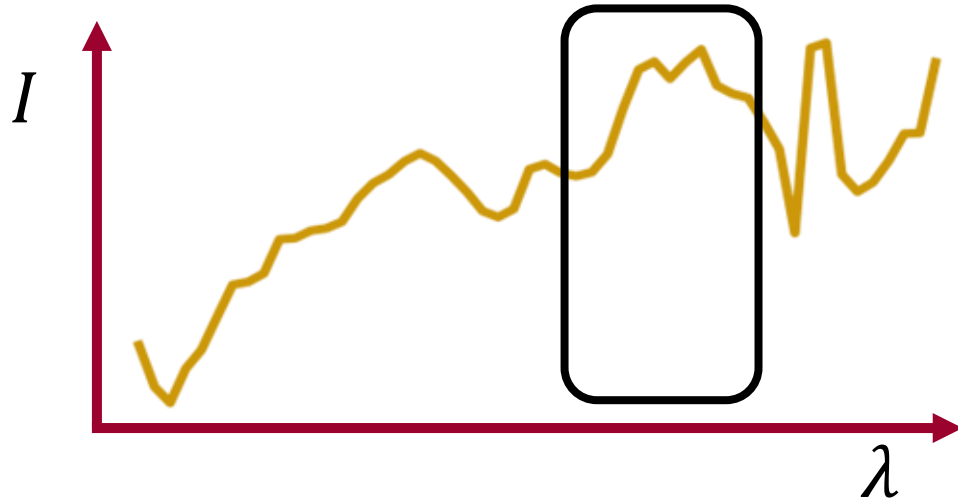
average rate of change in mean posterior
abundance retrieved by **TauREx-III**

Machine Learning Retrievals for Practical Application



Wish List

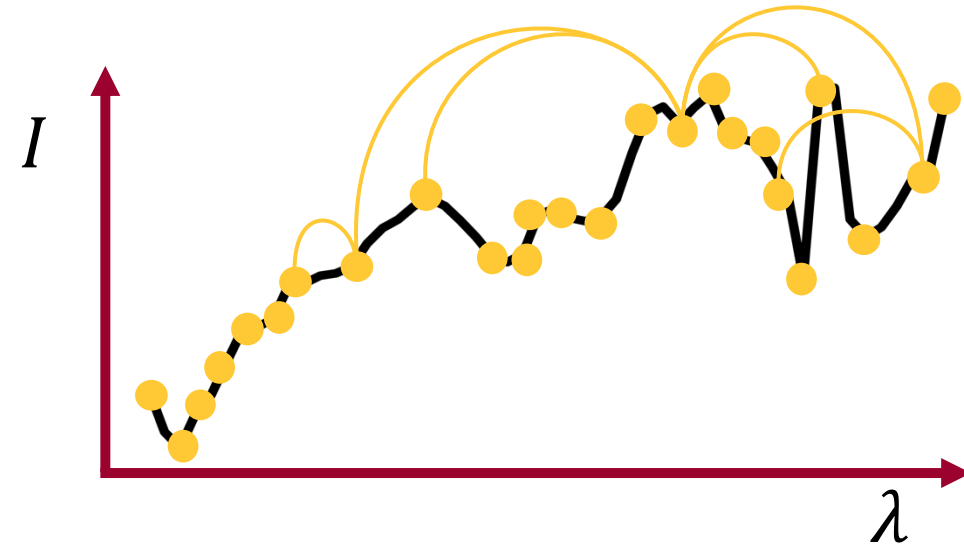
- ✓ *uncertainty quantification*
- ✓ *feature importance*
- ? *wavelength grid / instrument agnostic*
- ✗ *joint posterior*

CNN

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix}$$

One large
N-dimensional
datapoint

**sampling grid is baked into the
model as the true function**

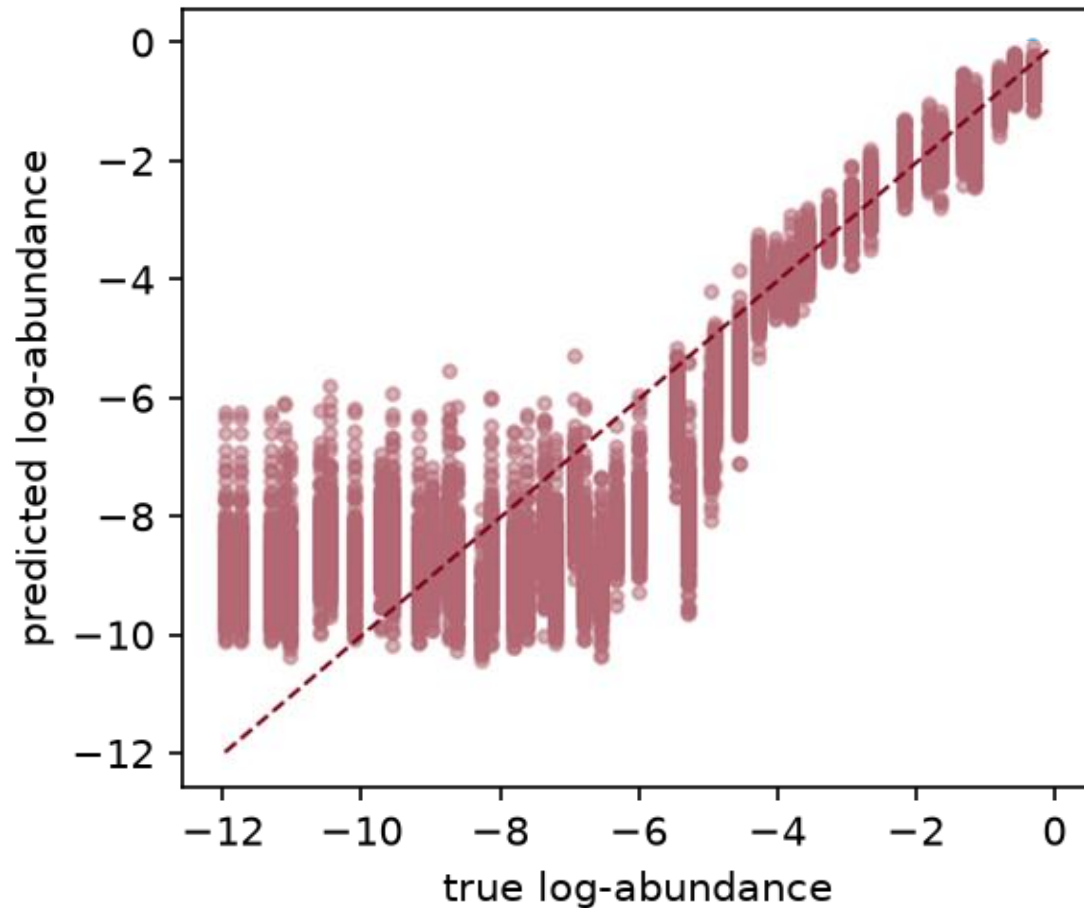
Set Transformer

$$\begin{bmatrix} I \\ \lambda \\ \text{width}_\lambda \end{bmatrix}$$

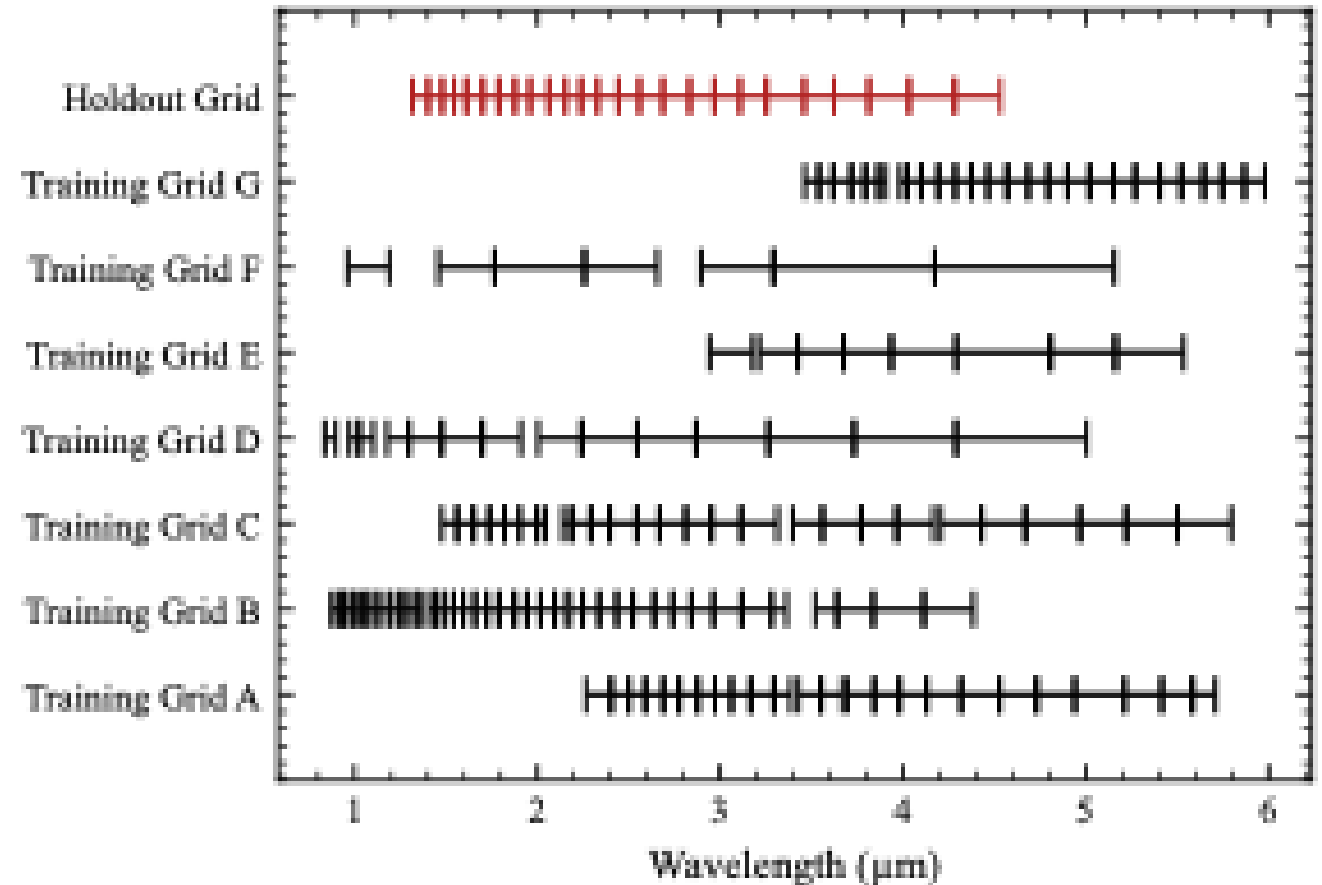
A set of N
3-dimensional
datapoints

**datapoints provide information on
an unknown true function**

Preliminary methane abundance retrievals for unseen wavelength grid.



Training across different data reductions and different instruments.



**Get in touch, get updates for this work,
and for any of my other research here!**



**I have less than a year left
on my PhD...**

